



Digital Communication IC Design

Lecture-6: Automatic Gain Control, Pulse Shaping and Receiving Filter

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Outline

- ☐ Iteration and Optimization
- ☐ Automatic Gain Control (AGC)
- ☐ Pulse Shaping and Receiving Filter

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Iteration and Optimization

- ❑ Three basic steps for solving an optimization problem
 - Step-1: Setting a goal – choosing a “**performance**” or “**objective**” function (as known as “**cost**” function)
 - Step-2: Choosing a method of achieving the goal – **minimizing** or **maximizing** the **objective** function
 - Step-3: Testing to make sure the method works as expected
- ❑ There are many ways to carry out a minimization or maximization problem.
 - For instance, if the problem is to find a point at which a polynomial function achieves its **minimum** value, this can be directly solved by finding the **derivative** (step-1) and setting it **equal** to zero (step-2).

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Iteration and Optimization (cont'd)

- ❑ However, the direct solutions are **impossible** in practical system.
- ❑ It is possible to find the optimized solution using **recursive** (or adaptive) approaches.
 - Steepest descent algorithm
 - $(\text{new guess}) = (\text{old guess}) + (\text{a correction term})$
$$w_{n+1} = w_n + \mu p, \quad n \geq 0$$
 - μ : a step-size ($0 < \mu < 1$)
 - p : an update direction scalar/vector
 - n : the number of iterations
- ❑ Finally, using a simulation to test and make sure the method works.

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An Example: Polynomial Minimization

- Goal: Find a value of x to make the polynomial $J(x) = x^2 - 4x + 4$ is minimum

- Plot cost function

- Quadratic form

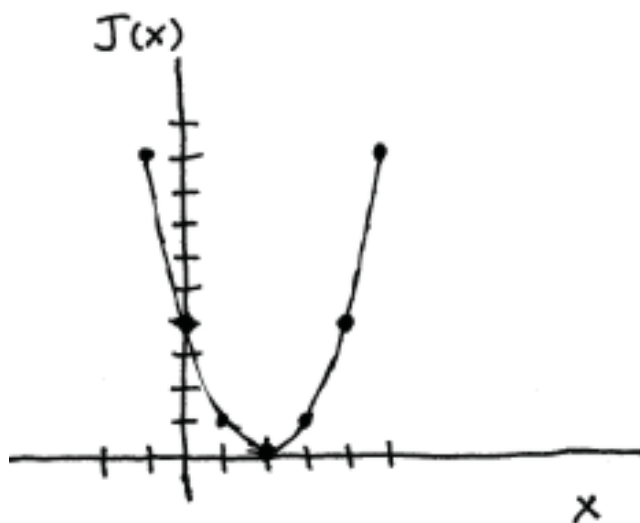
- Step-1: Take derivative

- $\frac{dJ(x)}{dx} = 2x - 4$

- Step-2: Set $\frac{dJ(x)}{dx} = 0$

- $2x - 4 = 0$

- Selecting $x = 2$ locates a stationary point, i.e., a **minimum of $J(x)$**



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Polynomial Minimization (cont'd)

- $J(x)$ minimum or maximum?

- The stationary point $x = 2$ is a minimum because the derivative of the derivative evaluated at $x = 2$ is positive.

- $\frac{d(2x-4)}{dx} = 2 > 0$

- A **negative second derivative** would indicate a **local maximum**.

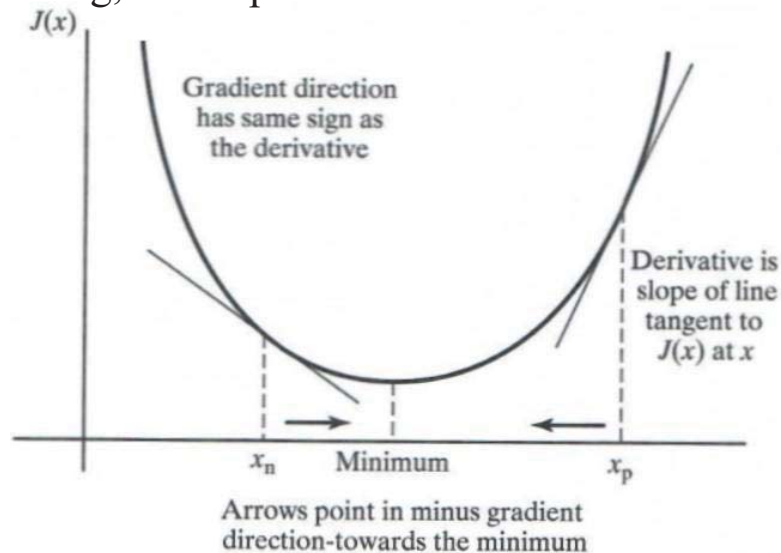
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Iterative Gradient Descent Minimizing Strategy

□ Adaptive method

- Steepest descent, which aims to minimize.
- Hill climbing, which points to maximize.



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Iterative Gradient Descent Minimizing Strategy (cont'd)

□ Steep descent

- Begin with an **initial guess** of the location of the **minimum**
- Evaluate which direction from this estimation is most **steeply "downhill"** and make a **new estimation** along the **"downhill"** direction

□ Hill climbing

- Begin with an **initial guess** of the location of the **maximum**
- Evaluate which direction **climbs** the most **rapidly** and make a **new estimation** along the **"uphill"** direction

□ The key points in this procedure

- The **uphill direction** is defined by the **gradient** evaluated at the **current** location (or the positive direction).
- The **downhill direction** is the **negative** of this **gradient**.

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Steepest Descent

□ To apply steepest descent to find the minimization of $J(x)$

- A current estimation of x at the time k , denoted $x[k]$
- A new estimation of x at the time $k + 1$, denoted $x[k + 1]$
- We have

$$x[k + 1] = x[k] + \mu \left\{ -\frac{d}{dx} J(x) \right\} \Big|_{x=x[k]}$$

- μ : the step-size (a small value, in general, $0 < \mu < 1.0$)
- $\frac{d}{dx} J(x) \Big|_{x=x[k]}$: the gradient of $J(x)$ at the current point $x[k]$

□ Go back to “Polynomial Minimization”

- $x[k + 1] = x[k] - \mu(2x[k] - 4)$
- $x[k + 1] = (1 - 2\mu)x[k] + 4\mu$

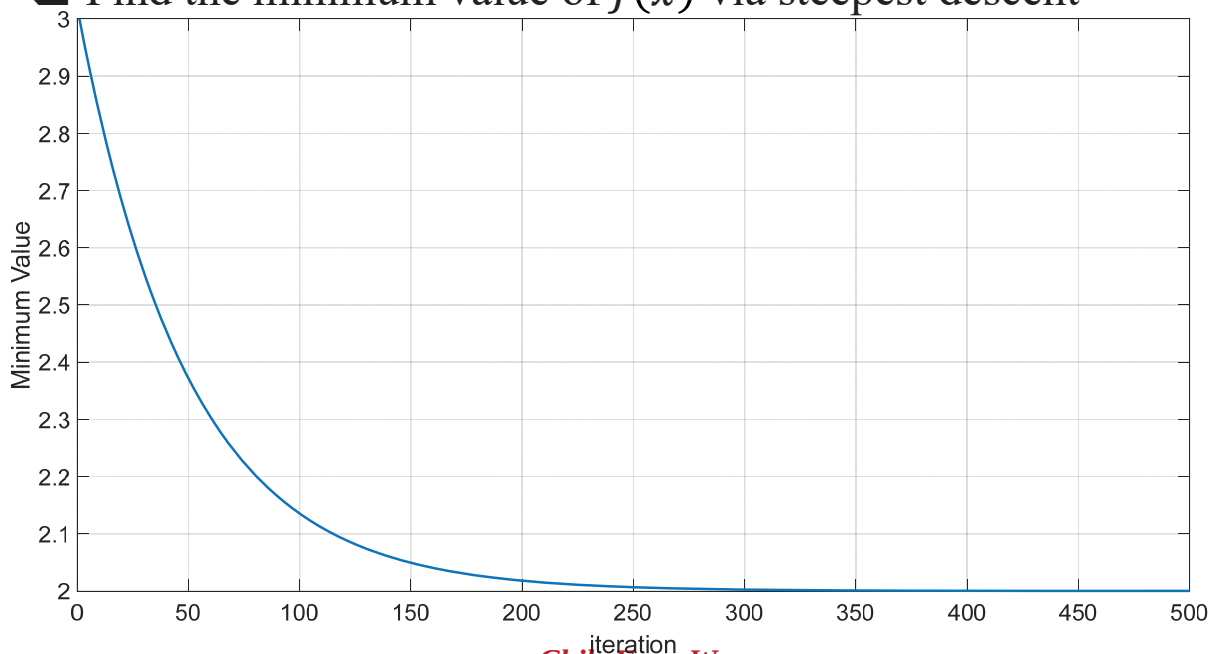
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Example

□ $N=500$ iterations; $\mu=0.01$; $x[1]=3$; $J(x) = x^2 - 4x + 4$

□ Find the minimum value of $J(x)$ via steepest descent

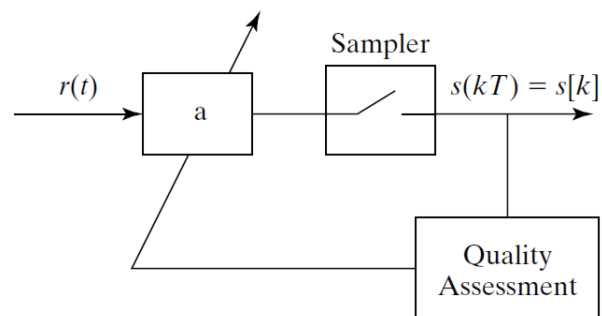


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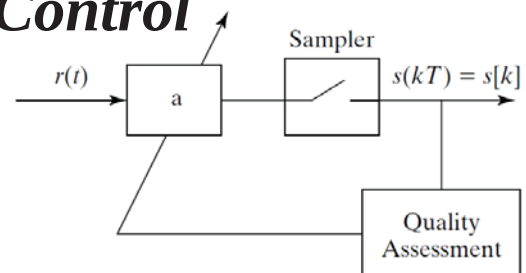
Application of Steepest Descent in AGC Design

- ❑ The AGC (Automatic Gain Control) provides a simple setting in all three of the major issues in optimization
 - Setting a goal
 - Choosing a method of solution
 - Verifying that the method is successful



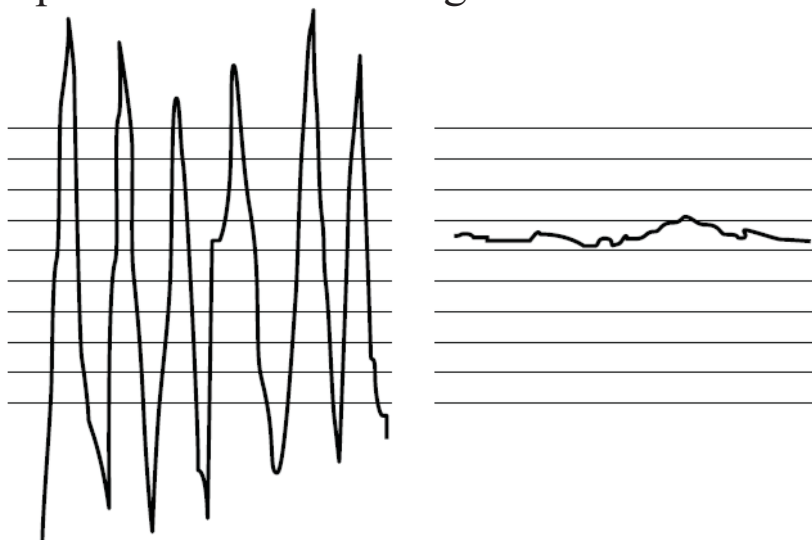
Automatic Gain Control

- ❑ The goal of AGC
 - **Amplify weak** signals,
 - **Attenuate strong** signals,
 - The signals remain within the normal operating range of the receiver (or the dynamic range of ADC at receiver)
- ❑ The power in a received signal depends on many things
 - The strength of the broadcast
 - The distance from the transmitter to the receiver
 - The direction in which the antenna is pointed
 - Any geographic features such as mountains (or building) that block, reflect, or absorb the signal



AGC (cont'd)

- ❑ An example for the received signals before ADC



- ❑ Maintain the dynamic range of a (zero-average) signal by attenuating when it is too large (left-side) and be amplifying when too small (right-side).

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Adjusting Approach

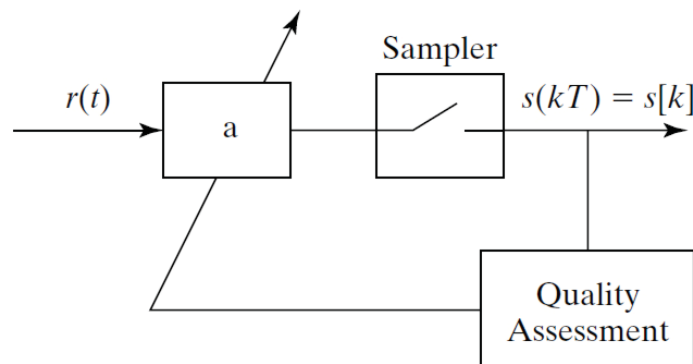
- ❑ Two basic approaches
 - Traditional approach: use analog circuit to adjust the gain before ADC
 - Adv.: the two blocks (the gain and the sampling) are separate and do NOT interact.
 - Modern approach: use the output of ADC to adjust the gain (in the front of ADC)
 - Adv.: The less additional hardware is required since the DSP processing is already present for other tasks.

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AGC Block

- A simple digital system for AGC gain adjustment



- $r(t)$: the input signal (the received signal)
- a : the gain to give the normalized signal $s(t)$
- $s[k]$: the sampled output
- Quality assessment: measure $s[k]$ and then determine whether a should be increased or decreased.

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AGC – Gain Tuning

- Goal and tuning

- Choose a so that the power (or average) of $s(t) \approx d^2$

$$[s(t)]^2 = a^2 \cdot \text{avg}\{r^2(t)\} \Big|_{t=kT} \approx \text{avg}\{s^2(kT)\} \approx \text{avg}\{s^2[k]\}$$
- It would be ideal to choose $a^2 \approx \frac{d^2}{\text{avg}\{r^2(kT)\}}$, because this would imply $\text{avg}\{s^2(kT)\} \approx d^2$.
- The averaging operation is defined as $\text{avg}\{x[k]\} = \frac{1}{N} \sum_{i=k-N+1}^k x[i]$.
 - A **moving average** over a data block of size N
- Unfortunately, the analog input $r(t)$ and its power $r^2(t)$ are NOT directly available to the assessment block in DSP portion of receiver.

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Adaptive Algorithm for Gain Tuning

□ Review of a viable optimization approach

- Setting a goal, choosing a solution method and testing

□ Goal – Approach-1 (Least-square)

- Minimize the difference between $s^2[k]$ and the desired power d^2
- The averaged squared error in $s^2[k]$ and d^2

$$J_{LS}(a) = \text{avg} \left\{ \frac{1}{4} (s^2[k] - d^2)^2 \right\}$$

$$= \frac{1}{4} \text{avg} \{ (a^2 \cdot r^2(kT) - d^2)^2 \}$$

- The parabolic form (quadratic form) in the polynomial minimization

Adaptive Algorithm for Gain Tuning (cont'd)

□ Applying the steepest descent strategy

- $a[k+1] = a[k] - \mu \left. \frac{dJ_{LS}(a)}{da} \right|_{a=a[k]}$

□ Derivation of $\frac{dJ_{LS}(a)}{da}$

$$\begin{aligned} \frac{dJ_{LS}(a)}{da} &= \frac{d}{da} \left\{ \frac{1}{4} \cdot \text{avg} \{ [a^2 \cdot r^2(kT) - d^2]^2 \} \right\} \\ &\approx \frac{1}{4} \cdot \text{avg} \left\{ \frac{d}{da} \{ [a^2 \cdot r^2(kT) - d^2]^2 \} \right\} \\ &= \text{avg} \{ [a^2 \cdot r^2(kT) - d^2] \cdot a \cdot r^2(kT) \} \\ &= \text{avg} \{ [s^2[k] - d^2] \cdot a \cdot r^2(kT) \} \quad \because a^2 \cdot r^2(kT) = s^2[k] \\ &= \text{avg} \left\{ [s^2[k] - d^2] \cdot \frac{s^2[k]}{a[k]} \right\} \quad \because a \cdot r^2(kT) = \frac{s^2[k]}{a} \end{aligned}$$

Adaptive Algorithm for Gain Tuning (cont'd)

□ The steepest descent algorithm

$$\blacksquare a[k + 1] = a[k] - \mu \cdot \text{avg} \left\{ (s^2[k] - d^2) \cdot \frac{s^2[k]}{a[k]} \right\}$$

Adaptive Algorithm for Gain Tuning (cont'd)

□ Goal – Approach-2 (Naive)

$$\blacksquare J_N(a) = \text{avg} \left\{ |a| \left(\frac{s^2[k]}{3} - d^2 \right) \right\} = \text{avg} \left\{ |a| \left(\frac{a^2 \cdot r^2(kT)}{3} - d^2 \right) \right\}$$

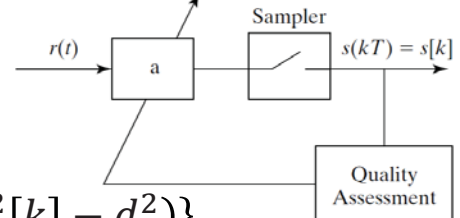
□ Derivation of $\frac{dJ_N(a)}{da}$

$$\begin{aligned} \frac{dJ_N(a)}{da} &= \frac{d}{da} \left\{ \text{avg} \left\{ |a| \left(\frac{a^2 \cdot r^2(kT)}{3} - d^2 \right) \right\} \right\} \\ &\approx \text{avg} \left\{ \frac{d}{da} \left\{ |a| \left(\frac{a^2 \cdot r^2(kT)}{3} - d^2 \right) \right\} \right\} \\ &= \text{avg} \left\{ \text{sgn}(a) \left(\frac{a^2 \cdot r^2(kT)}{3} - d^2 \right) + |a| \left(\frac{2a \cdot r^2(kT)}{3} \right) \right\} \\ &= \text{avg} \left\{ \text{sgn}(a) \left(\frac{a^2 \cdot r^2(kT)}{3} - d^2 \right) + \text{sgn}(a) \left(\frac{2a^2 \cdot r^2(kT)}{3} \right) \right\} \end{aligned}$$

Adaptive Algorithm for Gain Tuning (cont'd)

Derivation of $\frac{dJ_N(a)}{da}$

$$\begin{aligned}\frac{dJ_N(a)}{da} &= \text{avg} \left\{ \text{sgn}(a) \left(\frac{a^2 \cdot r^2(kT)}{3} - d^2 \right) + \text{sgn}(a) \left(\frac{2a^2 \cdot r^2(kT)}{3} \right) \right\} \\ &= \text{avg} \{ \text{sgn}(a) (a^2 \cdot r^2(kT) - d^2) \} \\ &= \text{avg} \{ \text{sgn}(a) (s^2[k] - d^2) \}\end{aligned}$$



The steepest descent algorithm

$$a[k+1] = a[k] - \mu \cdot \text{avg} \{ \text{sgn}(a[k]) (s^2[k] - d^2) \}$$

- Suppose "a" is positive and d is fixed,
- $\text{avg} \{ \text{sgn}(a[k]) (s^2[k] - d^2) \} = \text{avg} \{ (s^2[k] - d^2) \}$
 $= \text{avg} \{ s^2[k] \} - d^2$
- If $\text{avg} \{ s^2[k] \} > d^2$, then $a \downarrow$. If $\text{avg} \{ s^2[k] \} < d^2$, then $a \uparrow$.
- If $\text{avg} \{ s^2[k] \} \approx d^2$, then $a^2 \approx \frac{d^2}{r^2}$.

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Adaptive Algorithm for Gain Tuning (cont'd)

The steepest descent algorithm

$$a[k+1] = a[k] - \mu \cdot (\text{avg} \{ s^2[k] \} - d^2)$$

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Summary

❑ Cost function

- J_{LS} : minimize the **square** of **difference** between $s^2[k]$ and the desired power d^2
- J_N : minimize the **difference** between $s^2[k]$ and the desired power d^2

❑ Design parameter

- μ : a large $\mu \rightarrow$ allow $a[k]$ to change faster; a small $\mu \rightarrow$ allow greater smoothing;
- μ : a trade-off between the BW of $a[k]$ (or the convergent speed) and the amount of jitter (or noise)
- $avg\{\cdot\}$: the length (N) of averaging affecting the speed of adaptation
 - **Long** average: slower moving and smoother estimation
 - **Short** average: faster moving and jittery estimation

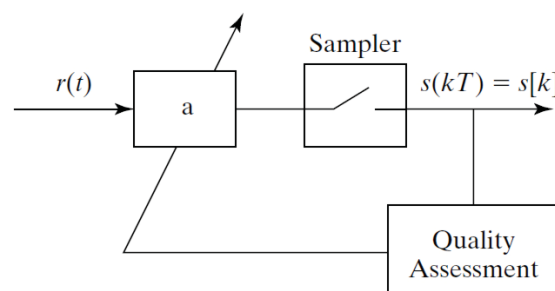
Example

- ❑ $d_s=0.15$, $avg\{r^2\} \approx 1$, iterations=10000, $a[1] = 1.0$ and N (the length of average)=10;

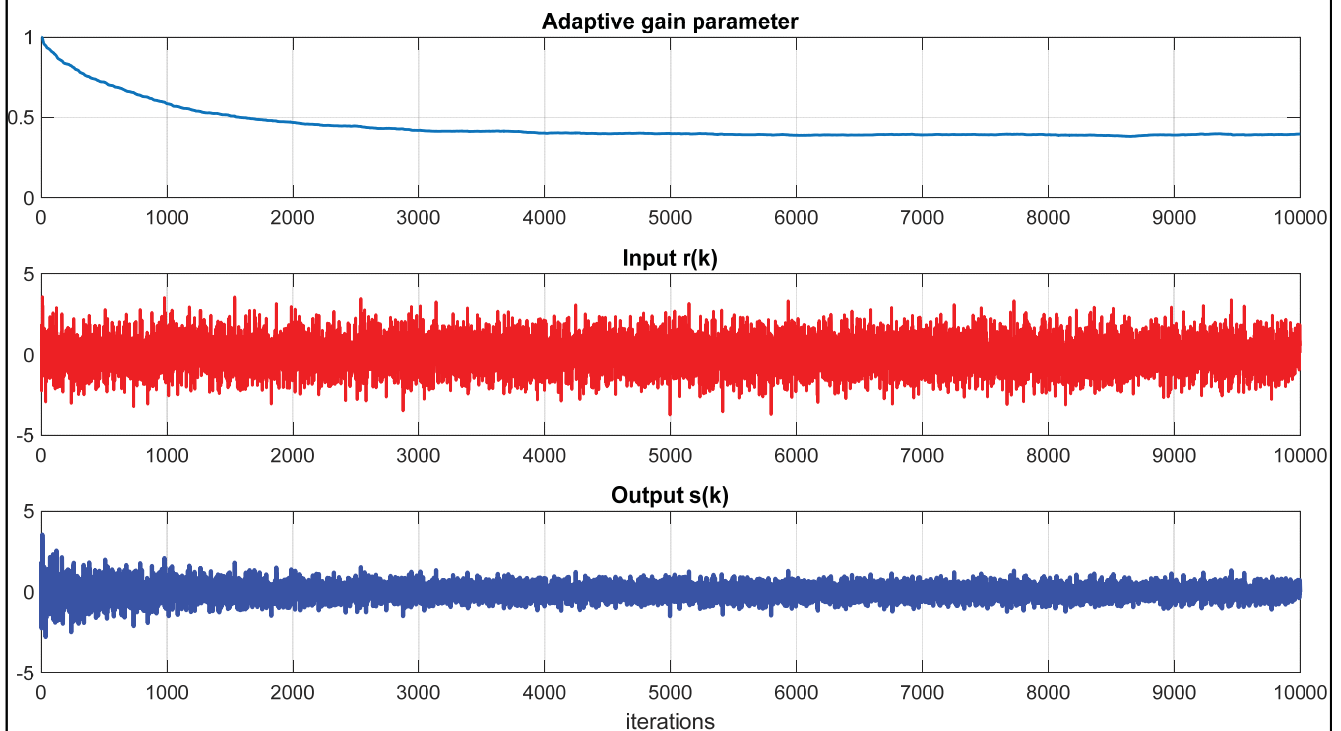
- Adjust the AGC to make the overall power of s roughly equal to the specified parameter d_s , which means that $s^2 \approx d_s \approx 0.15$

❑ Final results

- $a[10000] \approx 0.38 \rightarrow 0.38^2 \approx 0.15$



Example: Simulation Result



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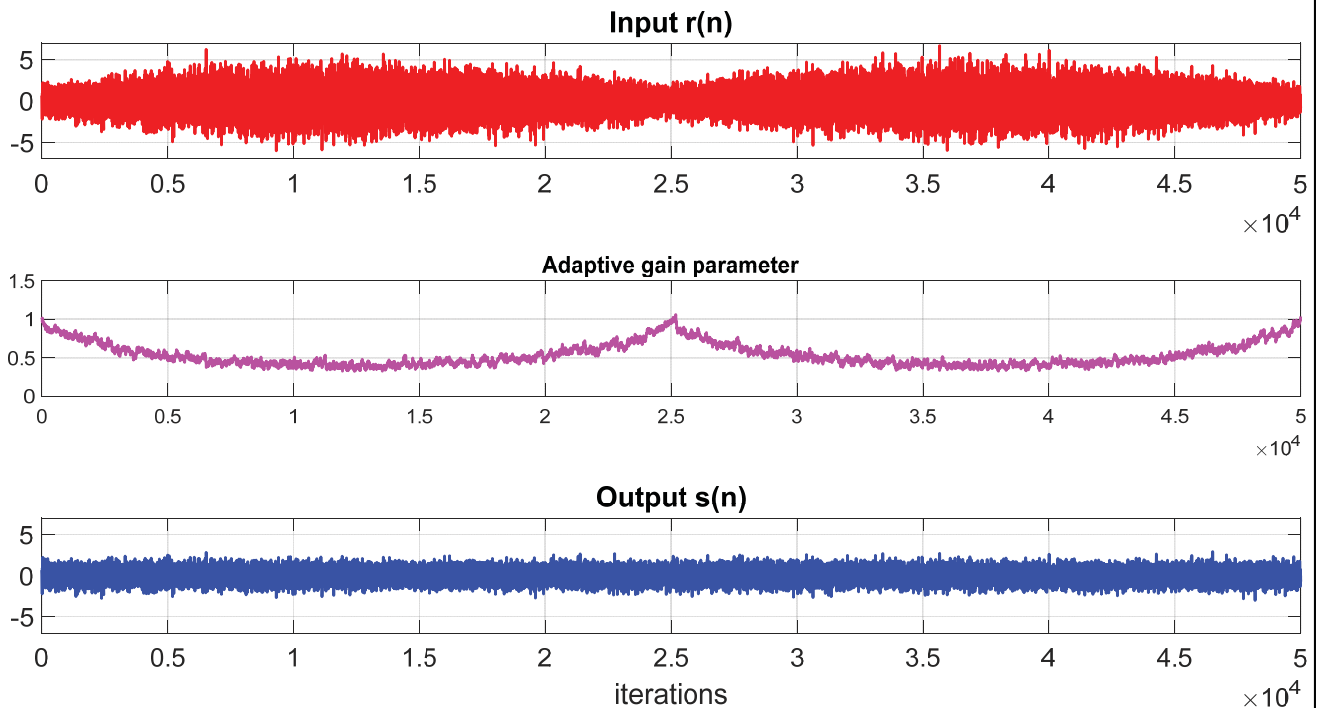
Using an AGC to Combat Fading

- ❑ One of the impairments encountered in transmission systems
 - The strength degradation of the received signal due to fading (or the transmission path)
- ❑ Show an AGC used to counteract the fading
 - Assume the slow fading and does NOT disappear of the signal in transmission

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Evaluation of Using AGC to Combat Fading

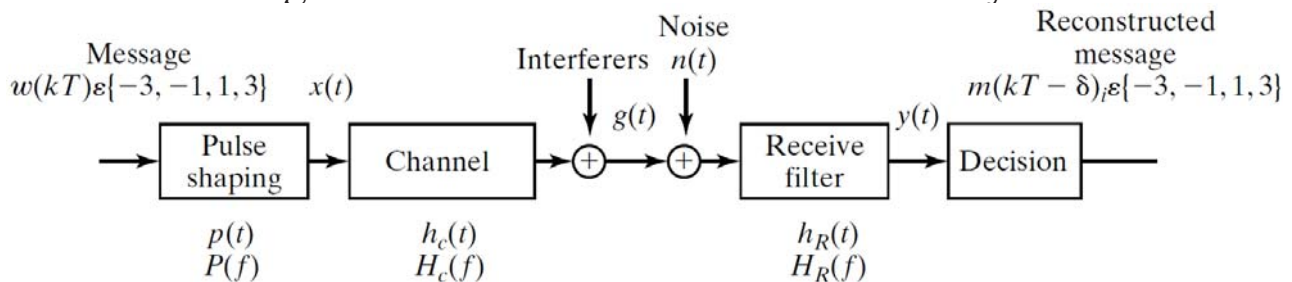


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Pulse Shaping and Receiving Filter

Block diagram of a baseband communication system



- Digital signal: The transmitted message and the reconstructed message
- Analog signal: The transmitted signal in channel and the received signal from channel
- Pulse shaping
 - Change each transmitted symbol in the digital message with a suitable analog pulse

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Pulse Shaping and Receiving Filter (cont'd)

❑ Receiving filter

- Assist in recapturing the digital values from the received pulses

❑ Information message

- $w(kT)$: a finite set of values
 - Indexed by k and the data rate of T seconds
- $m(kT)$: It is the same as $w(kT)$.

❑ Transmitted analog pulse

- Scaled by the value of the information message $w(kT)$

❑ Inter-symbol interference (ISI)

- The analog pulse is wider than the time between adjacent symbols and then the outputs of adjacent symbols may overlap.

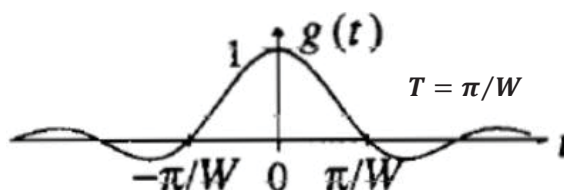
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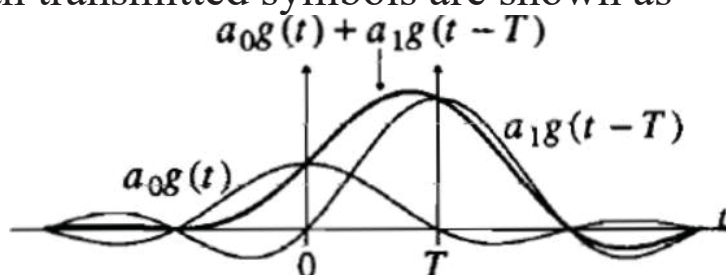
Pulse Shaping and Receiving Filter (cont'd)

❑ Ideal analog pulse

- $g(t) = \frac{\sin(Wt)}{Wt}$



- ❑ Considering two successive symbols with $a_0 = 1$ and $a_1 = 2$, the both transmitted symbols are shown as



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Spectrum for the Pulse

- ❑ The discrete-time message sequence is turned into the analog pulse

$$\omega_a(t) = \sum_k \omega(kT) \delta(t - kT) = \begin{cases} \omega(kT) & t = kT \\ 0 & t \neq kT \end{cases}$$

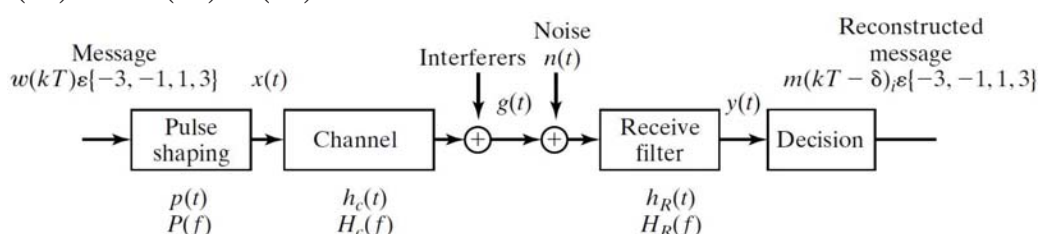
- The pulse shaping filter

- ❑ The response of the pulse shaping output is given as

$$x(t) = \omega_a(t) * p(t)$$

- ❑ The Fourier transform of $x(t)$

$$X(f) = W_a(f)P(f)$$



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Nyquist Pulse

- ❑ The corresponding output of the received filter is

$$y(t) = \omega_a(t) * p(t) * h_c(t) * h_R(t)$$

- h_c : the impulse response of the channel

- h_R : the impulse response of the received filter

- ❑ Let the equivalent impulse response

$$h_{equiv}(t) = p(t) * h_c(t) * h_R(t)$$

- ❑ The frequency response of the equivalent impulse response is

$$H_{equiv}(f) = P(f)H_c(f)H_R(f)$$

- ❑ No ISI effect

- The pulse shape is zero at all integer multiples of kT .

- One pulse does NOT interfere with other pulses at subsequent T -spaced samples instants.

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Nyquist Pulse (cont'd)

□ Nyquist pulse is defined as

$$h_{\text{NYQ}}(kT + \tau) = \begin{cases} c, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

- c : nonzero constant
- τ : the timing offset

□ Consideration of the Nyquist pulse

- Rectangular pulse (Time-width $< T$)
 - Too wide in frequency (BW)
- Requirement of the Nyquist pulse
 - Wide in time and narrow in frequency

Nyquist Pulse (cont'd)

□ **sinc** pulse (the narrowest possible spectrum)

$$h_{\text{sinc}}(t) = \frac{\sin(\pi f_0 t)}{\pi f_0 t}$$

- $f_0 = 1/T$
- A rectangle in frequency (i.e., a lowpass filter)

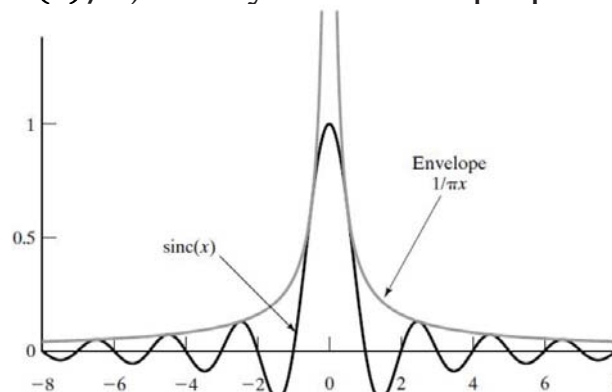
□ Assuming the clocks at the transmitter and receiver are synchronized, the $\tau = 0$.

- The **sinc** pulse is Nyquist because $h_{\text{sinc}}(0) = 1$ and

$$h_{\text{sinc}}(kT) = \frac{\sin(\pi k)}{\pi k} = 0 \text{ for all } k \neq 0$$

Nyquist Pulse (cont'd)

- ❑ There are some problems with the **sinc** pulse:
 - Infinite duration (truncation for real implementation)
 - Noncausal (delay (shift) for real implementation)
 - Difficult to approximate the steep band edges of the rectangular frequency function $H_{\text{sinc}}(f)$
 - Decay ($\text{sinc}(t)/t$) slowly with a rate proportional to $1/t$



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Nyquist Pulse (cont'd)

- ❑ Trade-off time and frequency behaviors
 - Narrower in time and a little wider in frequency
- ❑ Desirable pulse shapes
 - Appropriate zero crossings (i.e., Nyquist pulse)
 - Sloped band edges in the frequency domain
 - Decay more rapidly in the time domain (compared with **sinc**), while maintaining a narrow profile in the frequency domain

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Raised Cosine Filter

- Raised cosine-rolloff (or raised cosine) filter

$$H_{RC}(f) = \begin{cases} 1, & |f| < f_1 \\ \frac{1}{2} \left(1 + \cos \left[\frac{\pi(|f| - f_1)}{2f_\Delta} \right] \right), & f_1 < |f| < B \\ 0, & |f| > B \end{cases}$$

where

B : the absolute BW

f_0 : the 6 dB BW ($= 1/2T$, one half the symbol rate)

$f_\Delta = B - f_0$ and $f_1 = f_0 - f_\Delta$

- The corresponding time domain pulse

$$h_{RC}(t) = F^{-1} \{ H_{RC}(f) \} = 2f_0 \left(\frac{\sin(2\pi f_0 t)}{2\pi f_0 t} \right) \left[\frac{\cos(2\pi f_\Delta t)}{1 - (4f_\Delta t)^2} \right]$$

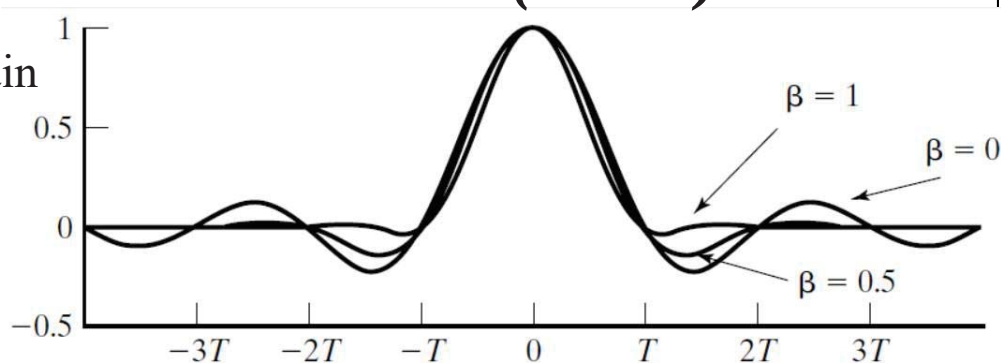
■ Rolloff factor $\beta = f_\Delta / f_0$

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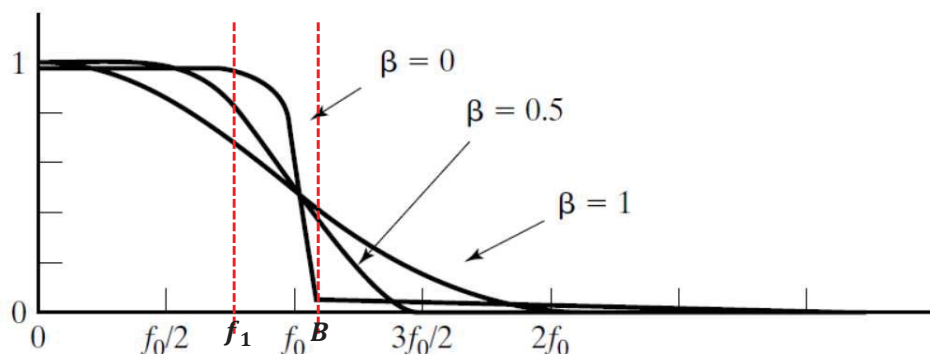
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Raised Cosine Filter (cont'd)

- Time domain pulse



- Frequency domain response

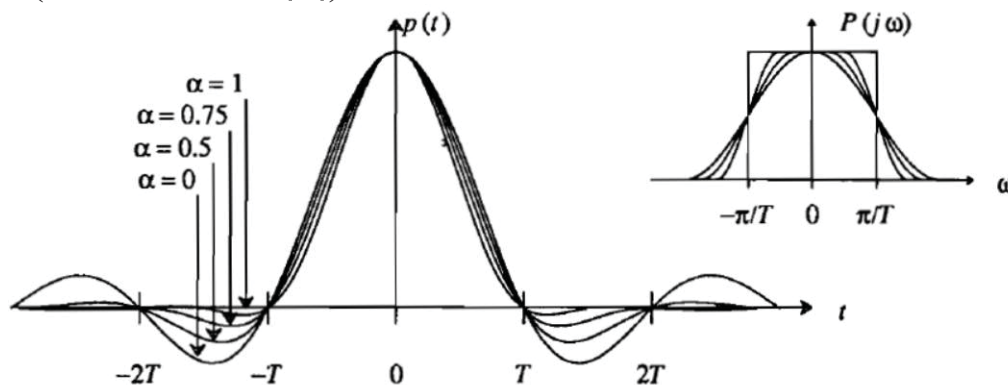


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Raised Cosine Filter (cont'd)

- The raised cosine pulse with nonzero β has the following characteristics:
 - Zero crossings at desired times
 - Less severe band edges $H_{RC}(f)$ compared with a **sinc** pulse
 - An envelope that falls off at approximately $1/|t|^3$ for large t (faster than $1/|t|$)

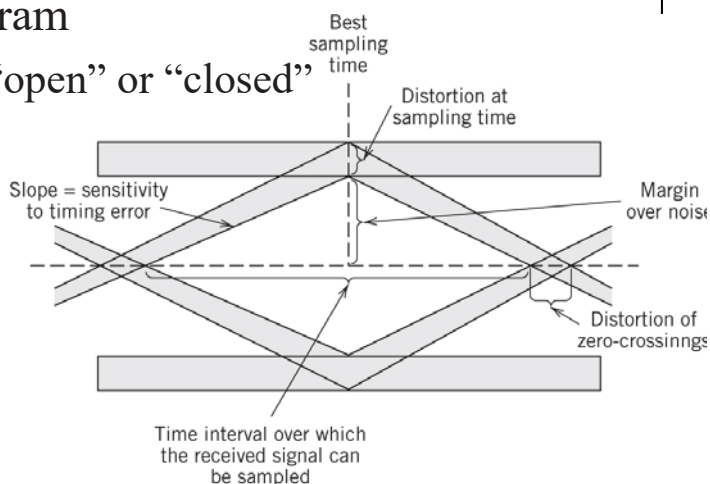


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Eye Diagrams

- Eye diagram (pattern)
 - An insightful manner to evaluate the ISI effect
 - Defined as the synchronized superposition of all possible received signal (receiver output)
- Interpretation of the eye diagram
 - Interior region to show eye “open” or “closed”
 - Time interval
 - Sensitivity to timing error
 - Distortion at sampling time
 - Best sampling time
 - Margin over noise
 - Distortion of zero-crossings



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Eye Diagrams

□ PAM-4 with $n=2$ symbols

□ Blue

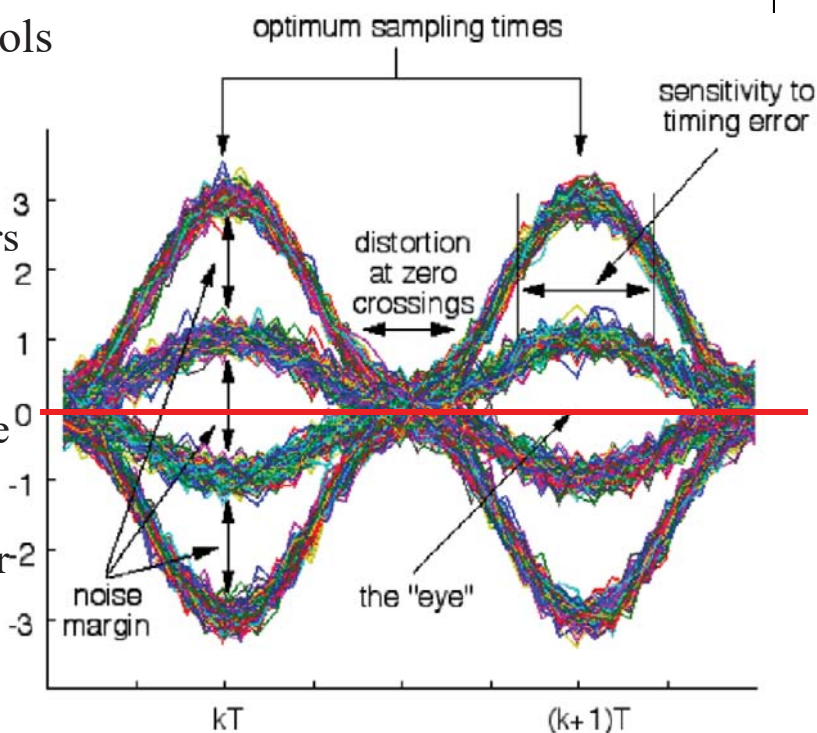
■ The immunity to errors in the timing phase

□ Green

■ The immunity to noise

□ Purple

■ The sensitivity to jitter in the timing phase

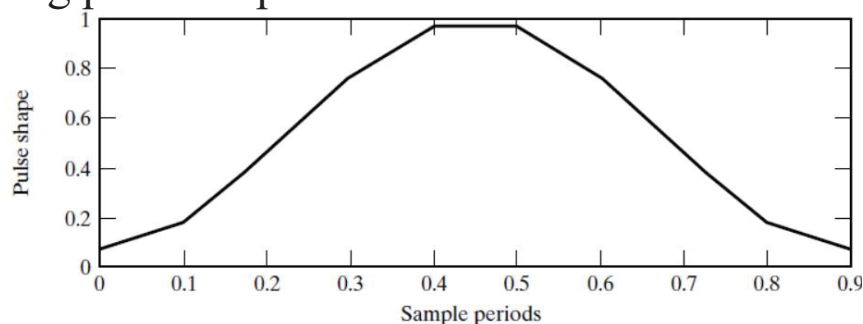


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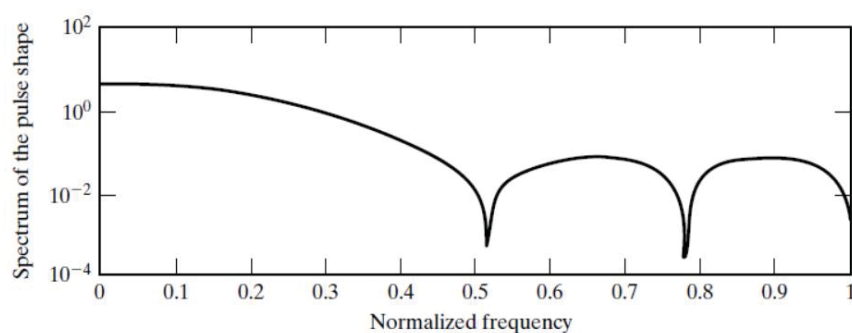
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Some Examples for Other Pulses

□ Hamming pulse shape



(a)

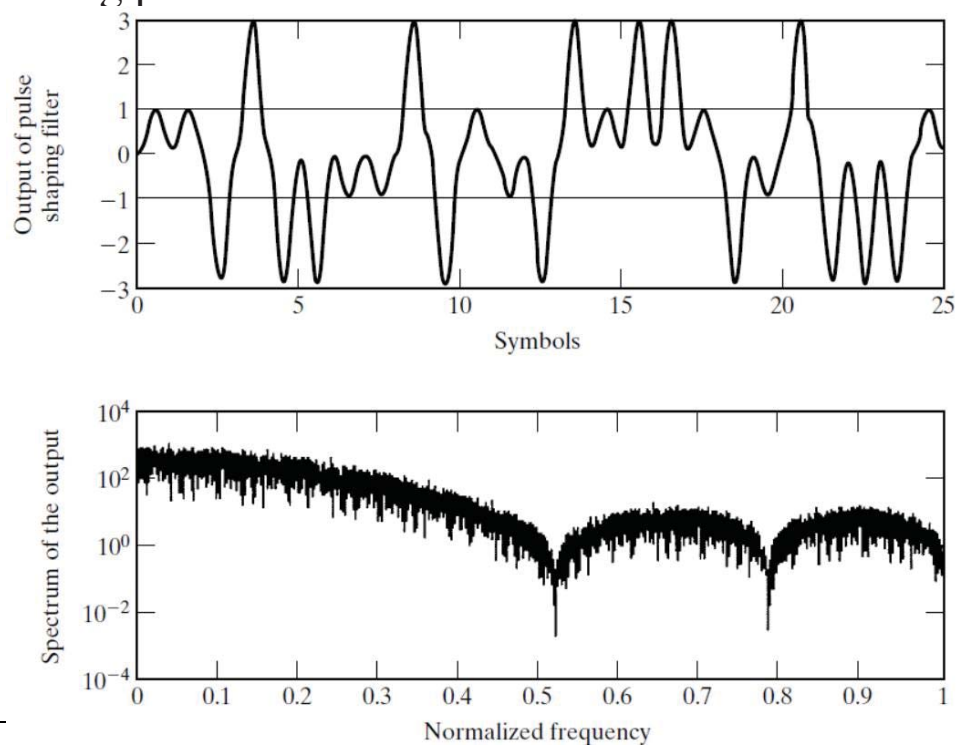


(b)

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Some Examples for Other Pulses (cont'd)

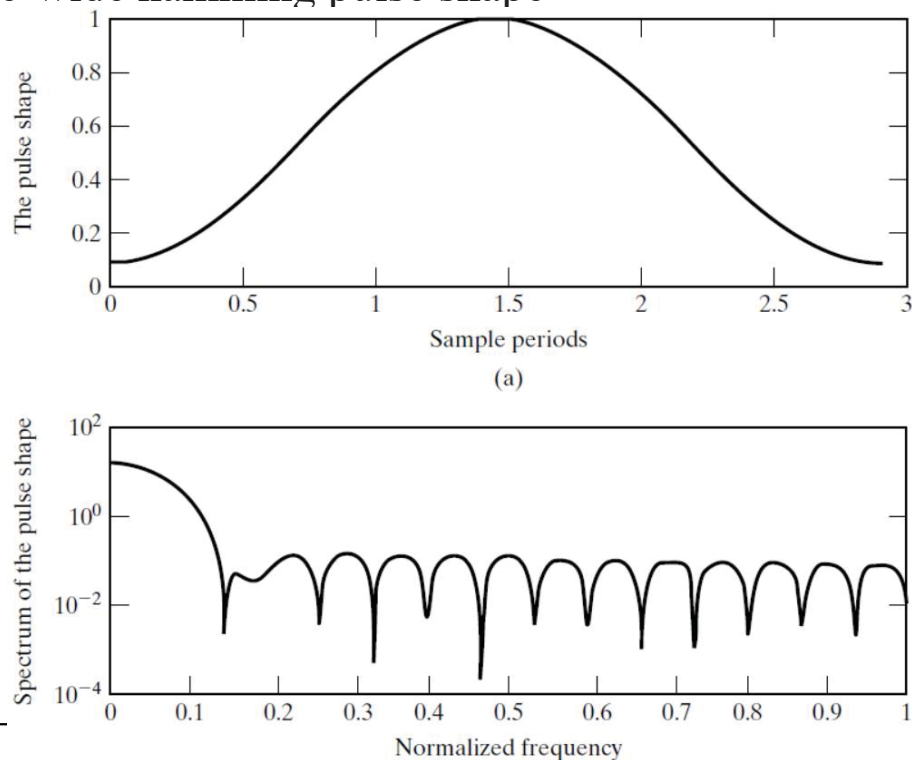
□ Hamming pulse with PAM-4



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Some Examples for Other Pulses (cont'd)

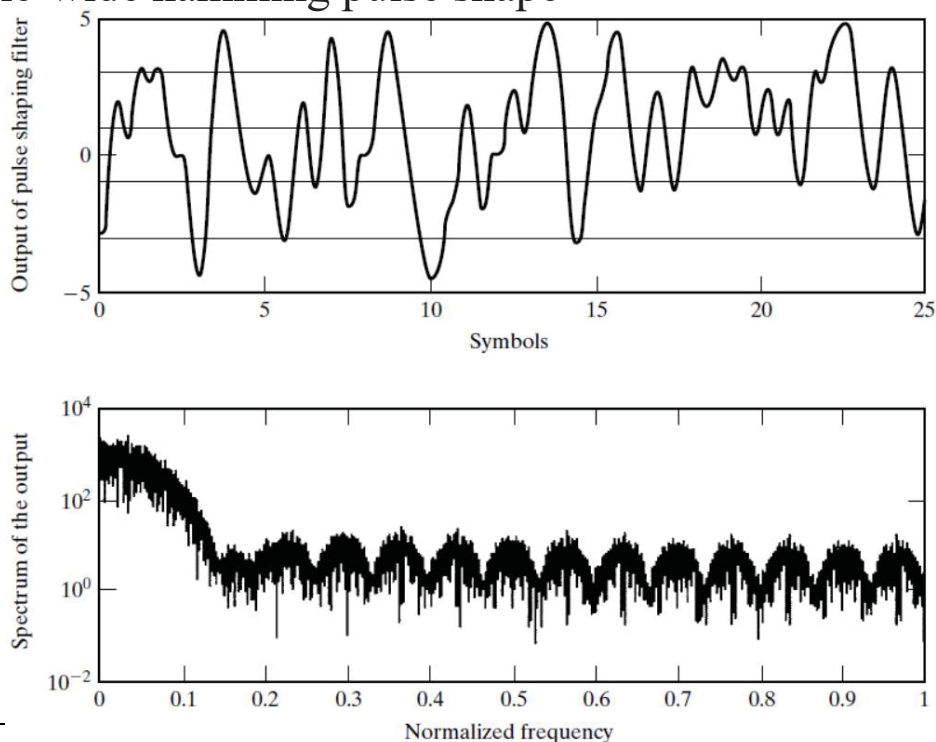
□ Triple-wide hamming pulse shape



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Some Examples for Other Pulses (cont'd)

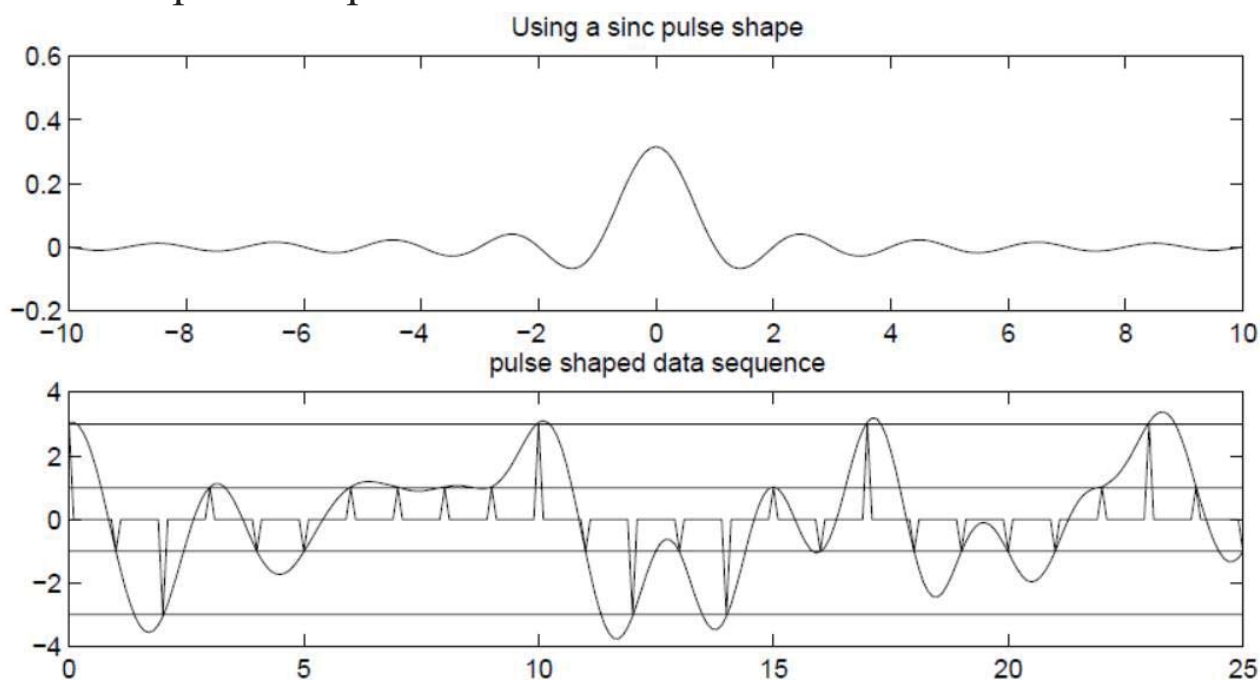
□ Triple-wide hamming pulse shape



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Some Examples for Other Pulses (cont'd)

□ sinc pulse shape with PAM-4

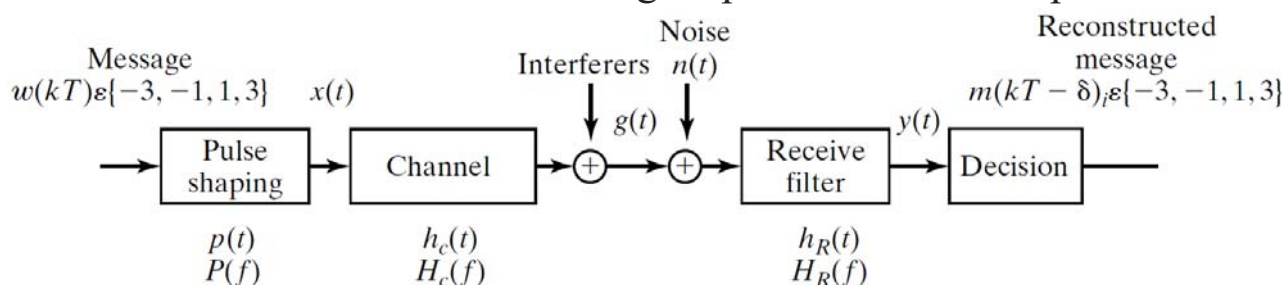


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Matched Filtering

□ Aim of matched filter

- Reduce the sensitivity to noise
- Maximize the ratio of the signal power to the noise power

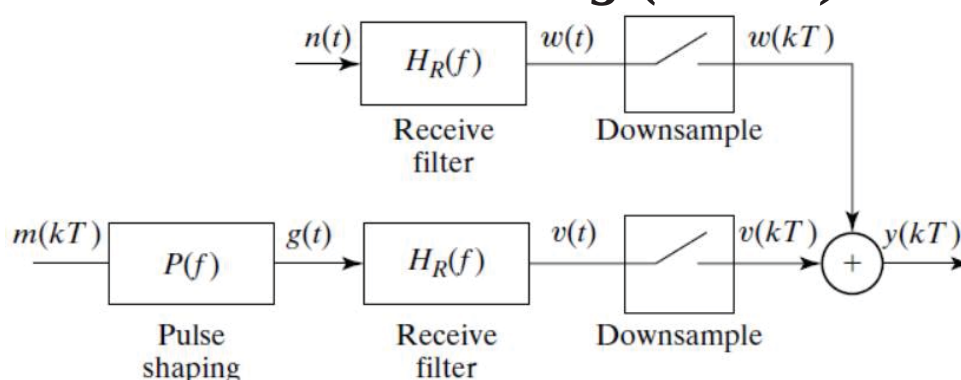


□ The output $y(t)$ is composed of $g(t)$ and $n(t)$, given as

$$y(t) = v(t) + z(t)$$

$$v(t) = h_R(t) * g(t) \text{ and } z(t) = h_R(t) * n(t)$$

Matched Filtering (cont'd)



□ Maximize the ratio of the power in the signal $v(kT)$ to the power in the noise $z(kT)$

- Choose $h_R(t)$ to maximize the power of the signal $v(t)$ at time $t = \tau$ compared with the power in $z(t)$
- Emphasize the signal $v(t)$ and suppress the noise $z(t)$



Matched Filtering (cont'd)

- ☐ Design an appropriate pair of filters
 - A pulse shape for the transmitter
 - A receiver filter that is matched to the pulse shape
- ☐ Nyquist pulse condition
 - The combination of the pulse shape and the received filter
- ☐ The received filter should have the following characteristics.
 - Allow no ISI at the receiver
 - Maximize the signal-to-noise ratio (SNR)



Matched Filtering (cont'd)

- ☐ Square-root raised cosine (SRRC) filter
 - The square-root is taken in the frequency domain, not the time domain.
 - The square of the SRRC is a Nyquist pulse.
- ☐ The time domain description of the SRRC pulse is found by taking the inverse Fourier transform of the square root of the spectrum of the raised cosine pulse.

Design of Square-Root Raised-Cosine Filter

□ Raised cosine spectrum

$$H_{RC}(f) = \begin{cases} T, & 0 \leq |f| < \frac{1-\beta}{2T} \\ \frac{T}{2} \left(1 + \cos \left[\frac{\pi T}{\beta} \left(|f| - \frac{1-\beta}{2T} \right) \right] \right), & \frac{1-\beta}{2T} \leq |f| < \frac{1+\beta}{2T} \\ 0, & |f| \geq \frac{1+\beta}{2T} \end{cases}$$

□ The time pulse of raised cosine filter

$$h_{RC}(t) = \frac{\sin(\pi t/T)}{\pi t/T} \left[\frac{\cos(\pi \beta t/T)}{1 - 4\beta^2 t^2/T^2} \right] = \text{sinc}(\pi t/T) \left[\frac{\cos(\pi \beta t/T)}{1 - 4\beta^2 t^2/T^2} \right]$$

Design of Square-Root Raised-Cosine Filter (cont'd)

□ Square-root raised-cosine filter

- The square-root of the raised cosine spectrum $\sqrt{H_{RC}(f)}$
- The time domain pulse of square-root raised cosine filter

$$h_{SRRC}(t) = \frac{\sin\left(\frac{\pi t}{T}(1-\beta)\right) + 4\beta \frac{t}{T} \cos\left(\frac{\pi t}{T}(1+\beta)\right)}{\frac{\pi t}{T} \left(1 - \left(4\beta \frac{t}{T} \right)^2 \right)}$$

- The $h_{SRRC}(t)$ is non-causal.
- Therefore, it has to be truncated, and the pulse shaping filter is typically realized with $\pm 6T$ around $t=0$.

Nyquist Pulse

- ❑ An ideal bandlimited channel W

$$H(w) = \begin{cases} 1, & |w| < W \\ 0, & |w| \geq W \end{cases}$$

- ❑ The ideally bandlimited pulse

$$G(w) = \begin{cases} \pi/W, & |w| < W \\ 0, & |w| \geq W \end{cases}$$

- Its time domain pulse is sinc pulse as $g(t) = \frac{\sin(Wt)}{Wt}$.

- ❑ Raised-Cosine Pulse